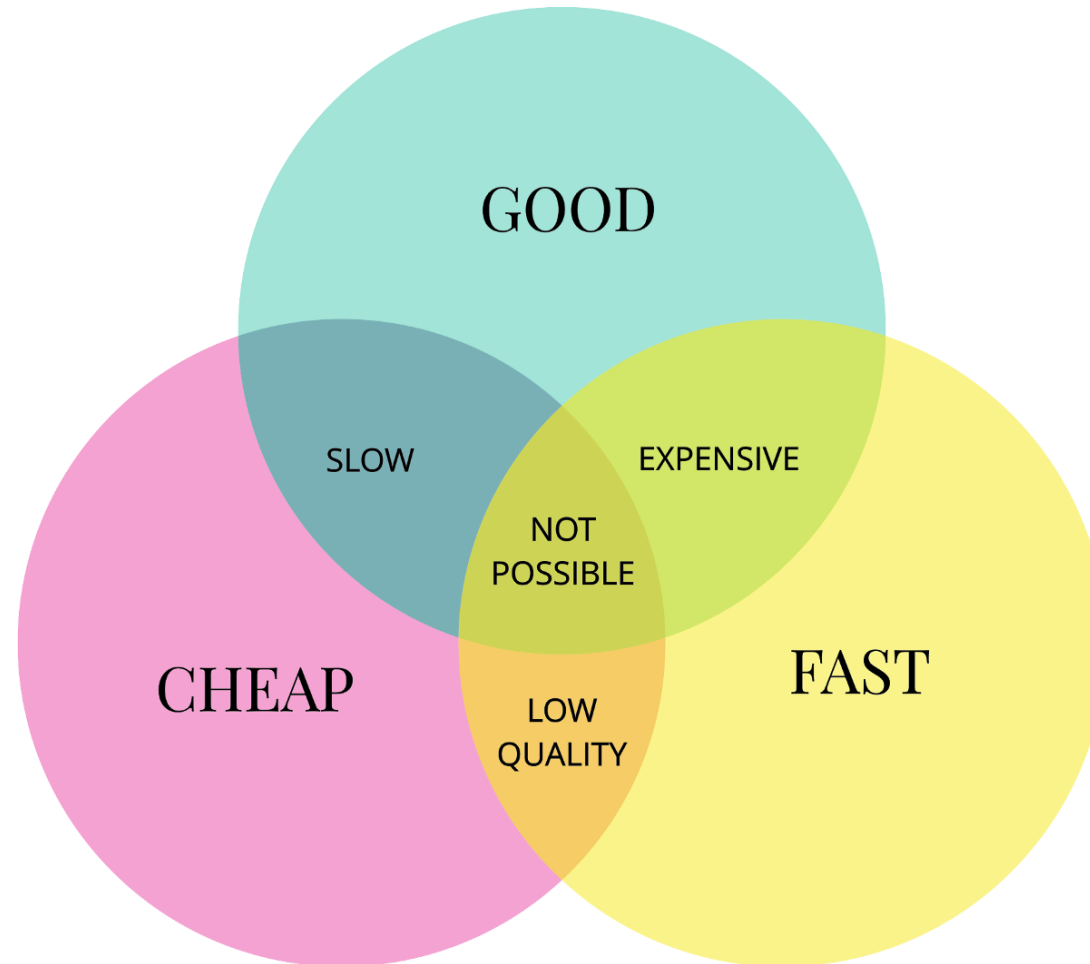


Requirements+

Other UP **requirement artifacts**
in addition to functional requirements

Fast, Cheap, Good: Choose any two!



Reliable specifications?

- Writing a Vision and Supplementary Specification is worthwhile as an exercise in clarifying a first approximation of what is wanted, the motivation for the product, and as a repository for the big ideas. But they are not – nor is any requirements artifact – a reliable specification. Only writing code, testing it, getting feedback, ongoing close collaboration with users and customers, and adapting, truly hit the mark.
- **This is not a call to abandon analysis and thinking, and just rush to code, but a suggestion to treat written requirements lightly, start programming early, and continually ideally, daily engage users and tests for feedback.**

Types of requirements

- **FURPS+**
 - **F**unctional – features, capabilities
 - **U**sability – human factors, help, documentation
 - **R**eliability – frequency of failures, recoverability, predictability
 - **P**erformance – response times, throughput, accuracy
 - **S**upportability – testability, adaptability, maintainability, compatibility, configurability, installability, scalability, localizability, and so on.
 - **+** everything else - resource limitations, languages and tools, hardware, licensing, etc.

Requirement types

- Requirements
 - capabilities and conditions to which the system must conform
- Functional requirements
 - Actions that the system must be able to perform
 - Responses to a user input
- Non-functional requirements
 - Usability, performance, security, availability, modifiability, portability, legal, etc.
 - They are related to the system rather than to single functions
- The non-functional requirements may be more critical than functional requirements.
 - If they are not met, the system may be rendered useless

Requirements organised in UP artefacts

- Functional requirements:
 - Use-case model
 - Typical scenarios
 - Functional requirements
- Non-functional requirements
 - Supplementary specifications
 - URPS+

Other possible UP artefacts

- The **supplementary specification** captures and identifies other kinds of requirements, such as reports, documentation, packaging, supportability, licensing and so forth.
- The **glossary** captures terms and definitions; it can also play the role of a data dictionary.
- The **vision** summarizes the “vision” of the project – an executive summary. It serves to tersely communicate the big ideas.
- The **business rules** (or domain rules) capture long-living and spanning rules or policies, such as tax laws, that transcend one particular application.

Glossary

- Noteworthy terms and their definition
- Start the glossary early
 - Helps understand the concepts / terms used by the customer
 - Get to know the domain
 - Gain common understanding
 - The same concept / term usually have different meaning by different stakeholder
 - Composite terms explained (and not only atomic terms)
 - E.g. sales which includes other elements, such as date and location
 - For each term in glossary:
 - Name and aliases
 - Description
 - Relation to other terms (concepts)
 - Format / Range of values / validation rules

Vision

1. A common vision of new system
 - Short
 - Not describing requirements in detail (but as functions)
 - Summarize some of the information in the use case model
2. The Key High-Level Goals
 - Summarizes the goals and problems at a high level (higher than use cases)
 - May belong to one use case or span many.
 - E.g. “we need fault-tolerant sales processing”
3. Problems of the stakeholders
 - Understand the players and what they are concerned about in relation to the system

Vision

4. Summary of System Features

- Features are behavioral functions a system can do
- Observable – fulfills a stakeholder need
- The system does <feature x>

5. Feature list

- Major features
- Commonly a two-level hierarchy of system features

6. Users

- User group or personas

- *POS services:*
 - *sales capture*
 - *payment authorization*
 - ...
- *Inventory management:*
 - *automatic reordering*
 - ...



Definition

Features are behavioral functions a system can *do*. They should pass this linguistic test:

The system does <feature X>.

Vision and use case

- UP suggested sequence
 1. Write a brief first draft of the Vision
 2. Identify user goals and the supporting use cases by name
 3. Write some use cases in detail, and start the Supplementary Specification
 4. Refine the Vision

Supplementary Specification

- Captures other requirements, information, and constraints not easily captured in the use cases or glossary.
- Mainly non-functional requirements
- If specific to use case:
 - First mention briefly under the “Special Requirements” section in the fully dressed use case
 - Later move it to Supplementary Specification to keep all non-functional requirements in one place

Use Case Section	Comment
Use Case Name	Start with a verb.
Scope	The system under design.
Level	"user-goal" or "subfunction"
Primary Actor	Calls on the system to deliver its services.
Stakeholders and Interests	Who cares about this use case, and what do they want?
Preconditions	What must be true on start, <i>and</i> worth telling the reader?
Success Guarantee	What must be true on successful completion, <i>and</i> worth telling the reader.
Main Success Scenario	A typical, unconditional happy path scenario of success.
Extensions	Alternate scenarios of success or failure.
Special Requirements	Related non-functional requirements.
Technology and Data Variations List	varying I/O methods and data formats.
Frequency of Occurrence	Influences investigation, testing, and timing of implementation.
Miscellaneous	Such as open issues.

Supplementary Specification

- Possible content:
 - Functionality
 - If common across many use cases
 - E.g. logging and error handling
 - If it does not fit into the use case concepts
 - Usability
 - Reliability
 - Recoverability
 - Performance
- (continue on next page)

Supplementary Specification

Revision History

Version	Date	Description	Author
Inception draft	Jan 10, 2031	First draft. To be refined primarily during elaboration.	Craig Larman

Introduction

This document is the repository of all NextGen POS requirements not captured in the use cases.

Functionality

(Functionality common across many use cases)

Logging and Error Handling

Log all errors to persistent storage.

Pluggable Rules

At various scenario points of several use cases (to be defined) support the ability to customize the functionality of the system with a set of arbitrary rules that execute at that point or event.

Security

All usage requires user authentication.

Usability

Human Factors

The customer will be able to see a large-monitor display of the POS. Therefore:

- Text should be large and clear. Speed, ease, and error-free processing are paramount in sales processing, as the buyer wishes to leave quickly, or they perceive the purchasing experience (and seller) as less positive.
 - Avoid colors and graphics. The cashier is often looking at the customer or items, not the computer display. Therefore, signals and warnings should be conveyed with sound rather than only via graphics.
- ### Reliability
- ### Recoverability
- If there is failure to use external services (payment authorizer, accounting system, ...) try to solve with a local solution (e.g., store and forward) in order to still complete a sale. Much more analysis is needed here...
- ### Performance
- As mentioned under human factors, buyers want to complete sales processing very quickly. One bottleneck is external payment authorization. Our goal: authorization in less than 1 minute, 90% of the time.

Supplementary Specification (continued)

- Possible content continued:
 - Supportability
 - Adaptability – e.g. fit changing needs
 - Configurability
 - Implementation Constraints
 - Must use certain technologies
 - External components

Supportability

Adaptability

Different customers of the NextGen POS have unique business rule and processing needs while processing a sale. Therefore, at several defined points in the scenario (for example, when a new sale is initiated, when a new line item is added) pluggable business rule will be enabled.

Configurability

Different customers desire varying network configurations for their POS systems, such as thick versus thin clients, two-tier versus N-tier physical layers, and so forth. In addition, they desire the ability to modify these configurations, to reflect their changing business and performance needs. Therefore, the system will be somewhat configurable to reflect these needs. Much more analysis is needed in this area to discover the areas and degree of flexibility, and the effort to achieve it.

Implementation Constraints

NextGen leadership insists on a Java technologies solution, predicting this will improve long-term porting and supportability, in addition to ease of development.

Purchased Components

- Tax calculator. Must support pluggable calculators for different countries.

Free Open Source Components

In general, we recommend maximizing the use of free Java technology open source components on this project.

Although it is premature to definitively design and choose components, we suggest the following as likely candidates:

Supplementary Specification (continued)

- Possible content continued :
 - Hardware and Software Constraints
 - Hardware
 - SW – e.g. operating system, external collaborating systems
 - Implementation Constraints
 - Must use certain technologies

Interfaces

Noteworthy Hardware and Interfaces

- Touch screen monitor (this is perceived by operating systems as a regular monitor, and the touch gestures as mouse events)
- Barcode laser scanner (these normally attach to a special keyboard, and the scanned input is perceived in software as keystrokes)
- Receipt printer
- Credit/debit card reader
- Signature reader (but not in release 1)

Software Interfaces

For most external collaborating systems (tax calculator, accounting, inventory, ...) we need to be able to plug in varying systems and thus varying interfaces.

Supplementary Specification (continued)

- Possible content continued:
 - Application-Specific Domain (Business) Rules
 - E.g. discount rules in an order system
 - Information in Domain of interests
 - Rules, standards, etc. with relation to the system
 - Legal issues
 - Reports / statistics
 - Internalization concerns
 - Licensing

Application-Specific Domain (Business) Rules

(See the separate Business Rules document for general rules.)

ID	Rule	Changeability	Source
RULE1	Purchaser discount rules. Examples: Employee20% off. Preferred Customer10% off. Senior15% off.	High. Each retailer uses different rules.	Retailer policy.

Legal Issues

We recommend some open source components if their licensing restrictions can be resolved to allow resale of products that include open source software.

All tax rules must, by law, be applied during sales. Note that these can change frequently.

Information in Domains of Interest

Pricing

In addition to the pricing rules described in the domain rules section, note that products have an *original price*, and optionally a *permanent markdown price*. A product's price (before further discounts) is the permanent markdown price, if present. Organizations maintain the original price even if there is a permanent markdown price, for accounting and tax reasons.

Credit and Debit Payment Handling

When an electronic credit or debit payment is approved by a payment authorization service, they are responsible for paying the seller, not the buyer. Consequently, for each payment, the seller needs to record monies owing in their accounts receivable, from the authorization service. Usually on a nightly basis, the authorization service will perform an electronic funds transfer to the seller's account for the daily total owing, less a (small) per transaction fee that the service charges.

Sales Tax

Sales tax calculations can be very complex, and regularly change in response to legislation at all levels of government. Therefore, delegating tax calculations to third

Domain (Business) Rules

- Possible content:
 - How a domain or business may operate
 - Company policies
 - Government laws
 - Physical laws
 - Useful to identify and record domain rules in a separate application-independent artifact
 - Can be reused across project
 - Licensing

Application-Specific Domain (Business) Rules

(See the separate Business Rules document for general rules.)

ID	Rule	Changeability	Source
RULE1	Purchaser discount rules. Examples: Employee20% off. Preferred Customer10% off. Senior15% off.	High. Each retailer uses different rules.	Retailer policy.

Other UP requirement artefacts

UP

Discipline	Artifact	Incep.	Elab.	Const.	Trans.
	Iteration →	I1	E1..En	C1..Cn	T1..T2
Business Modeling	Domain Model		s		
Requirements	Use-Case Model	s	r		
	Vision	s	r		
	Supplementary Specification	s	r		
	Glossary	s	r		
	Business Rules	s	r		
Design	Design Model		s	r	
	SW Architecture Document		s		
	Data Model		s	r	

Exercises

- Solve exercises 1 to 3
 - Note 1: Do you already have a stakeholder analysis in your business part of the project? Maybe sum up the analysis and refer to it from this vision.
 - Note 2: We learn about analysing the user group thoroughly next session (personas ect.)